

November 5, 2021

Applications of DFS

We can answer structural questions about graphs using DFS.

Maybe the most basic structural question is about connectivity

For directed graphs we can define two kinds of connectivity

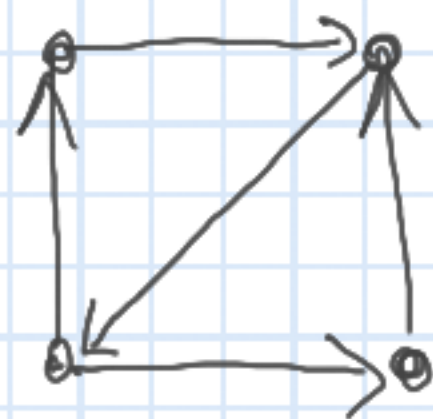
→ weak connectivity (we omit the directions of edges)

→ strong connectivity

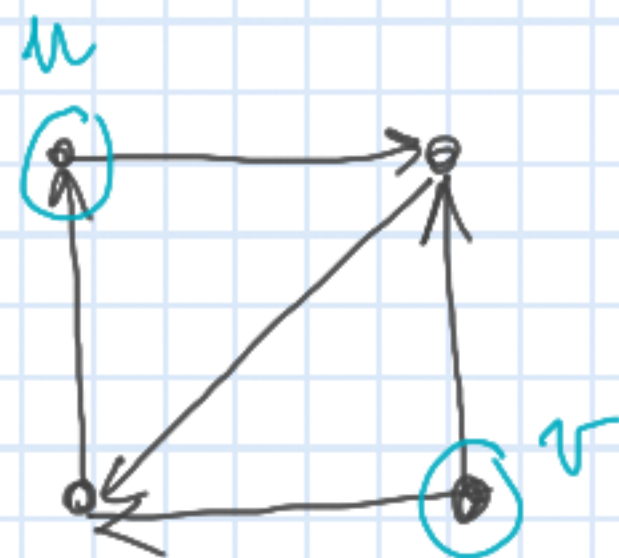
Definition

A directed graph $G = (V, E)$ is called strongly connected if for any two different vertices u and v there is a directed path from u to v as well as a directed path from v to u .

Example



strongly
connected graph



not strongly
connected
graph

there is no
directed path
from u to v

If a graph is not strongly connected we can also define the strongly connected components of the graph as maximal strongly connected subgraphs of it.

Definition

A subgraph H of a graph G is called a strongly connected component of G if

- (1) H is strongly connected
- (2) H is not a proper subgraph of any other strongly connected subgraph of G

Using DFS (twice) one can easily identify the strongly connected components of a directed graph $G = (V, E)$

- ① Run DFS on $G = (V, E)$, and record the finishing times of the vertices
- ② Produce the transpose $G^T = (V, E^T)$ of G (reverse the orientation of the edges)
- ③ Run DFS on $G^T = (V, E^T)$ in such a way that when you select a new starting

vertex, then this is the one whose finishing time is the largest one among the not yet discovered vertices.

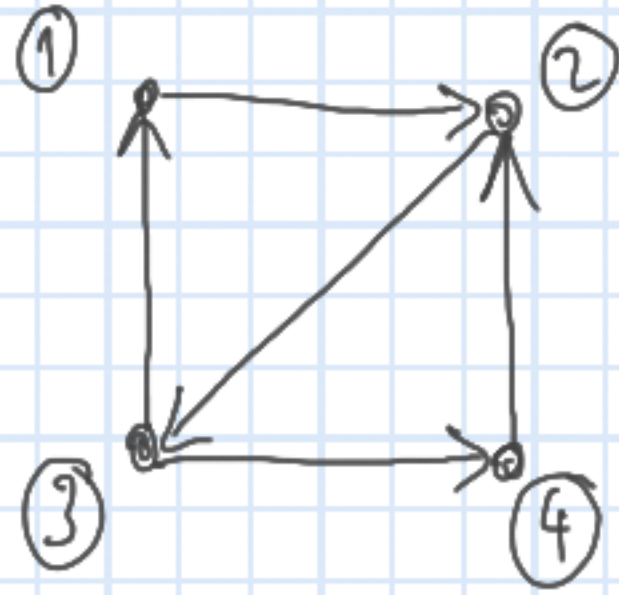
Cost

If the graph is given by adjacency lists, then the cost is $O(|V| + |E|)$

Remark

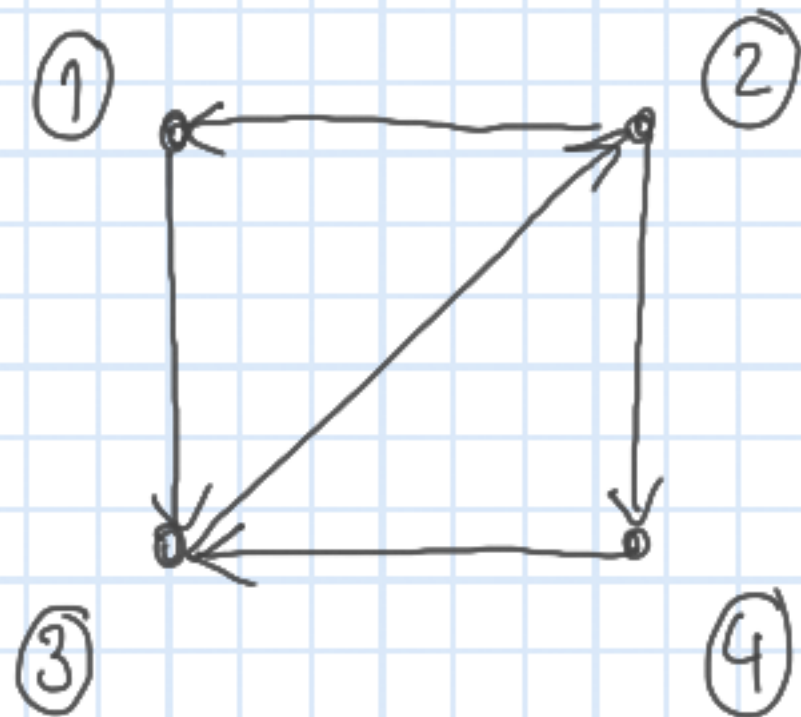
Producing $G^T = (V, E^T)$ from $G = (V, E)$

a.) G is given by an adjacency matrix



$$\text{Adj} : \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

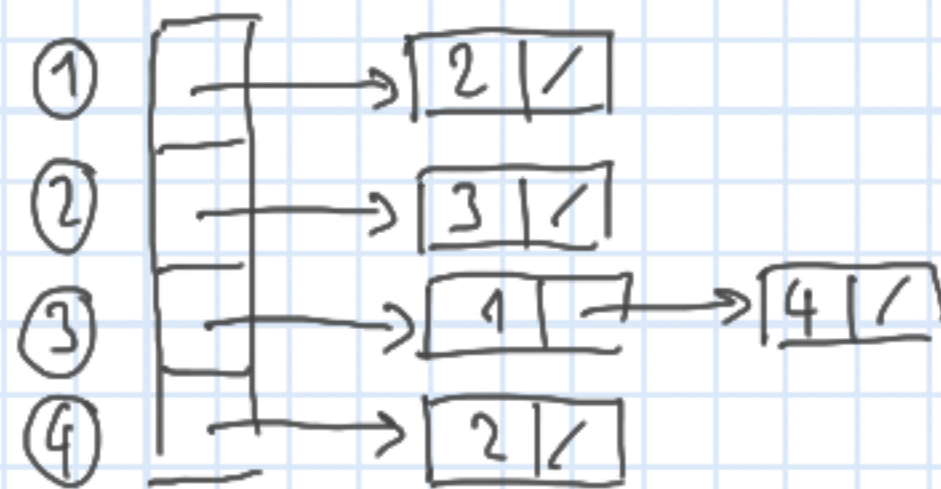
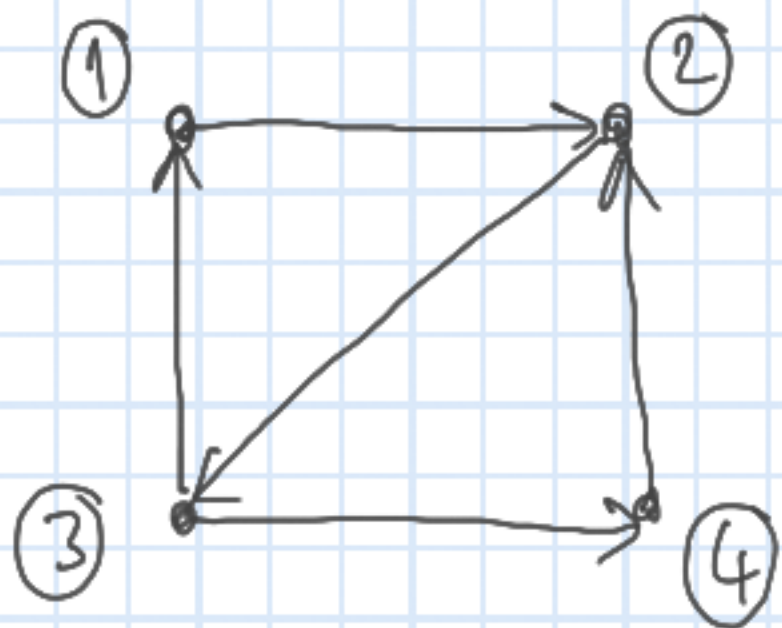
transpose



$$\text{Adj} : \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

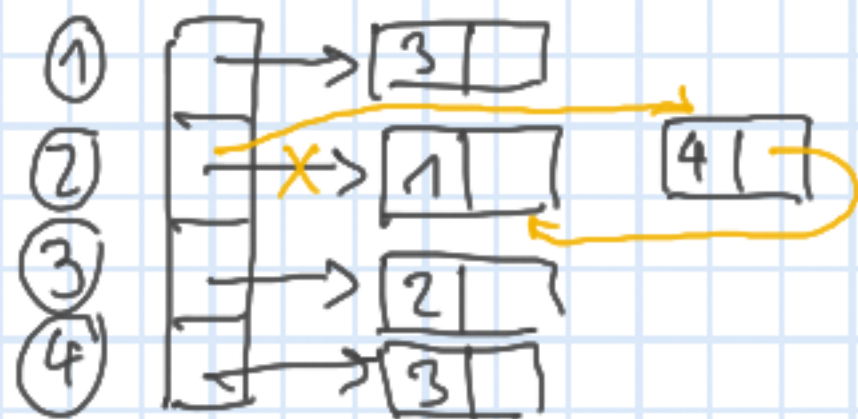
transpose of
the previous
one!

b., G is given by adjacency lists



How can we produce the adjacency list representation of G^T ?

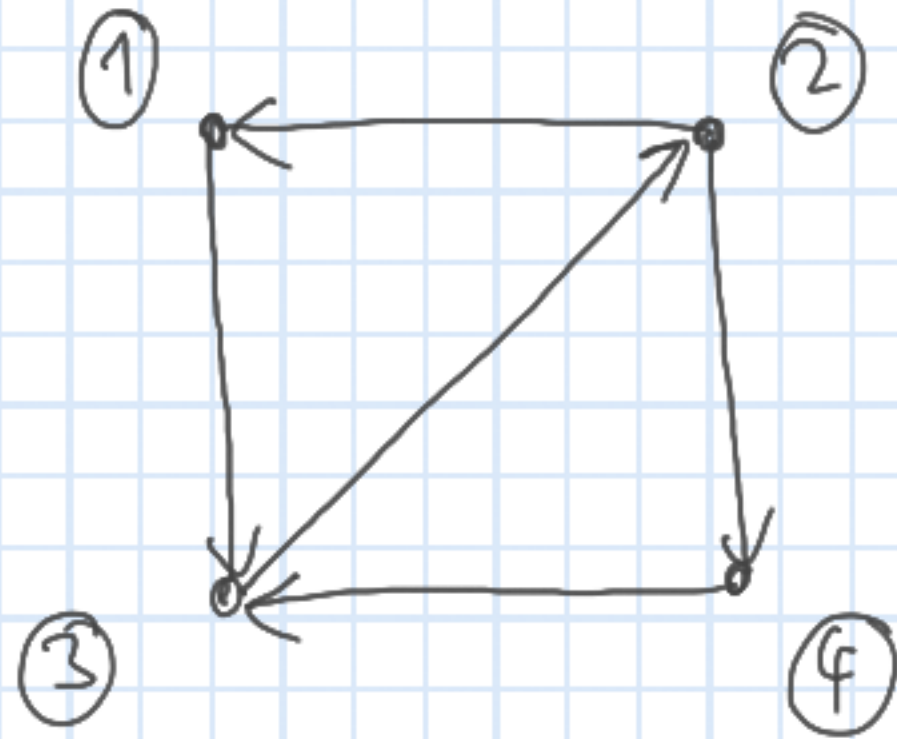
The vertex set remains the same



take each vertex v (one by one) and insert v into the adjacency lists of its neighbours

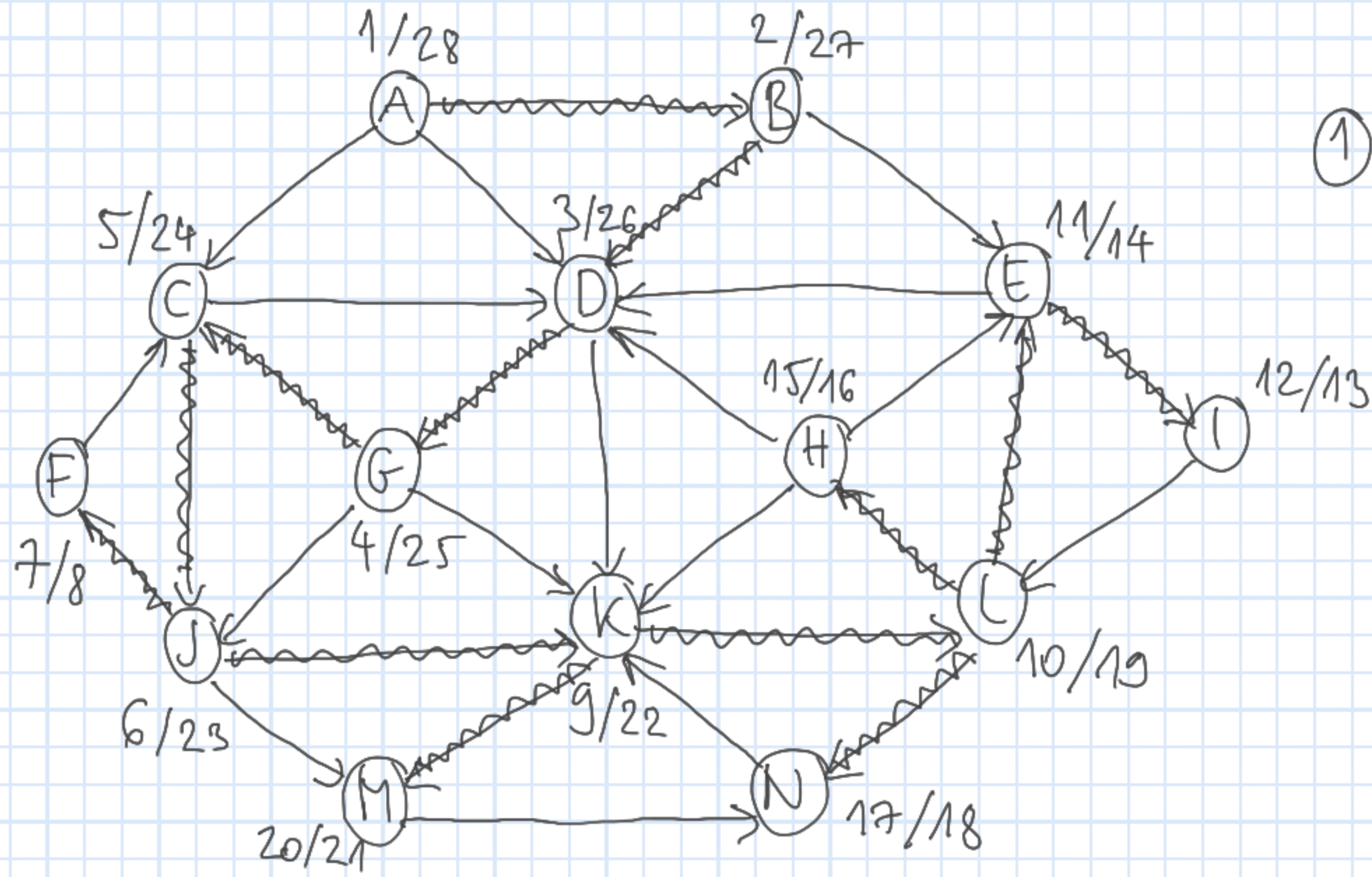
We insert into the beginning of the adjacency list to keep the insertion fast ($O(1)$ time)

In the previous example



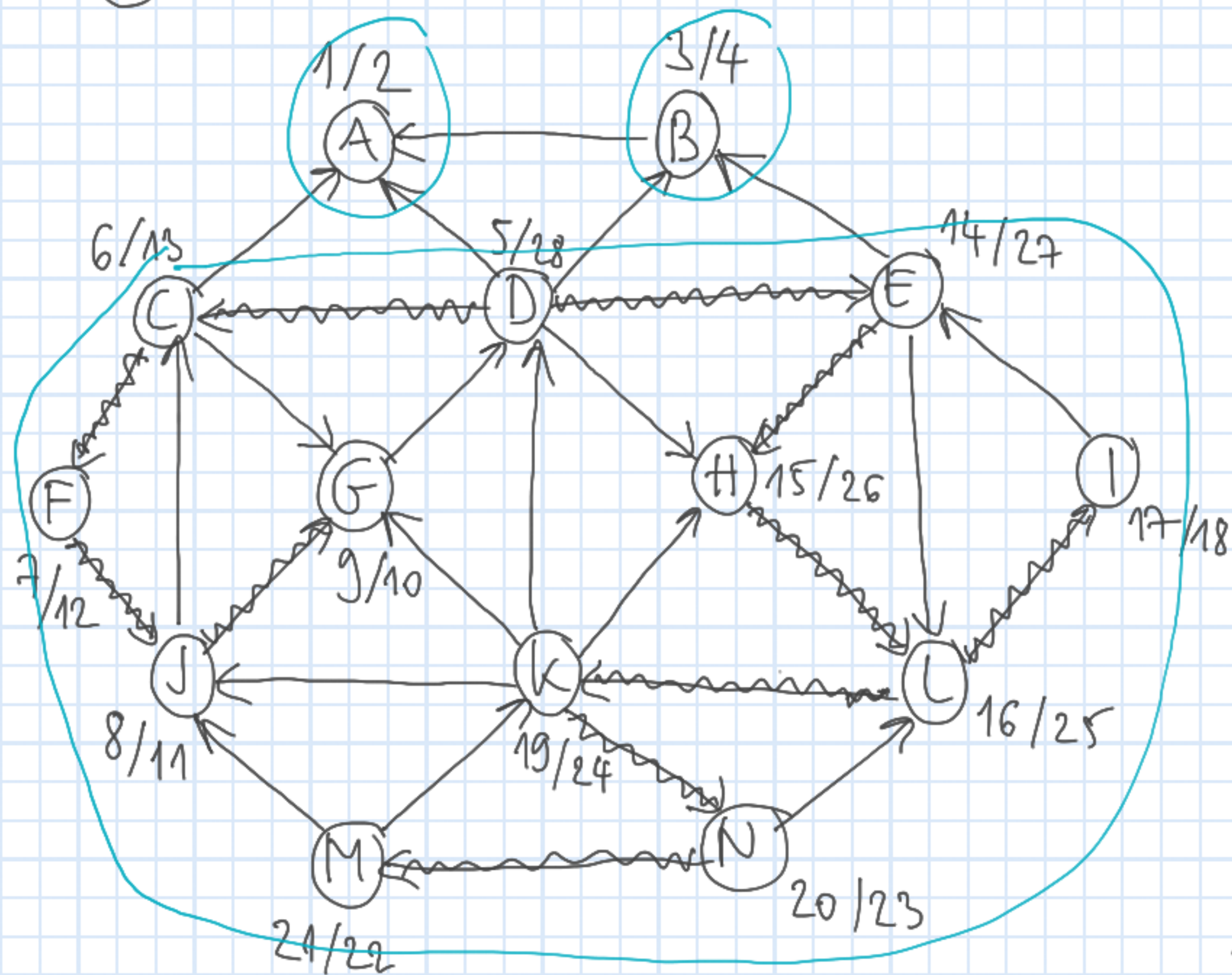
this is indeed the transpose

Find the strongly connected components of the "quiz graph" G :



① Run DFS on G

② Produce G^T

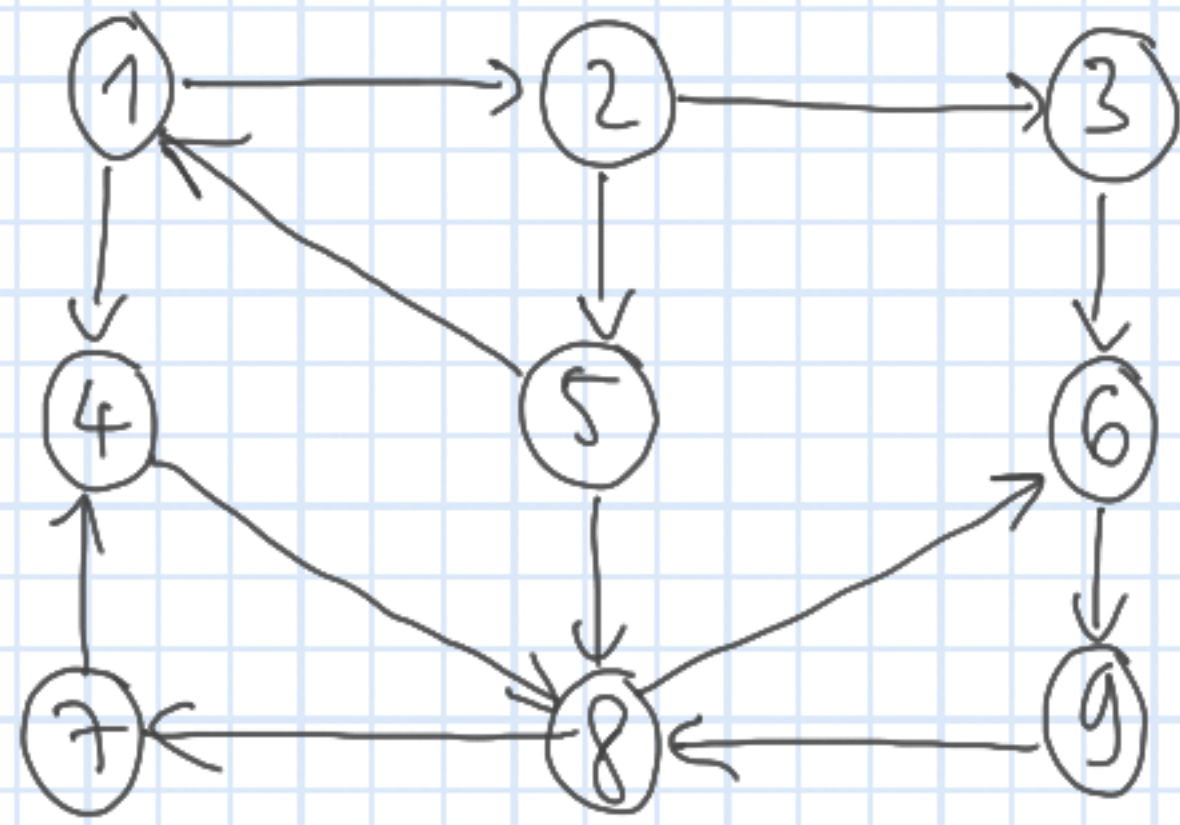


③ Run DFS on G^T in the described way

Strongly connected components : components of this DFS forest

We have three strongly connected components in G^T as well as in G (the vertex sets of the strongly connected components are the same in G and G^T)

Exercise (Quiz Problem)



Find the strongly connected components of this graph!

Soln

